Lab 8

Q1) Write an assembler code to reverse string X in order; use LODSB and STOSB strings;
where X DB ‘ HELLO WORLD’

X DB 'HELLO WORLD'
Y DB 11Dup(?)

```
ORG 100H

    MOV SI,OFFSET X[10]
    LEA DI,Y

    MOV CX,11

 L1:STD
    LODSB
    CLD
    STOSB
    LOOP L1

HLT

    X DB 'HELLO WORLD'
    Y DB 11 DUP(?)
```

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Q2) Write an assembler code to scan string X for * character ; replacing it by space.

X DB 'ABC*+GFDK**KLM'

and display it on the screen position (20, 40)

```
ORG 100H

    LEA DI,X
    CLD
    MOV CX,15
 L2:MOV AL,'*'
    SCASB
    JNZ L1
    MOV AL,20H
    DEC DI
    STOSB
 L1:LOOP L2

    MOV AH,02H
    MOV BH,00
    MOV DH,20
    MOV DL,40
    INT 10H

    MOV AH,09H
    LEA DX,X
    INT 21H

HLT

    X DB 'ABC*+GFDK**KLM'
```

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Q3) Write a sequence of instructions using (LODSB and STOSB) that copy each alphabet from X string to Y string, converting it to lowercase . Note: setting bit 5 in an uppercase character will convert it to lowercase

```
X   DB   'ABCDEFGHI'
Y   DB   LENGTHOF X DUP(' ')
```

Displaying the result on the screen.

```
ORG 100H

    MOV SI,OFFSET X
    MOV DI,OFFSET Y
    MOV CX,10
    CLD

 L1:LODSB
    OR AL,20H
    STOSB
    LOOP L1

    MOV AH,02H
    MOV BH,00
    MOV DH,0
    MOV DL,0
    INT 10H

    MOV AH,09H
    LEA DX,Y
    INT 21H

HLT

    X   DB   'ABCDEFGHI$'
    Y   DB    9 DUP('')
```

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